DO NOW — RECALL

Lesson 156 · Client-Side versus Server-Side Processing

Name: _____ Class: _____ Date: _____

1. Why does PageRank divide a page's score among outgoing links?

2. What data structure follows LIFO?

3. What is a database foreign key?

4. What is the role of a router?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Each outgoing link gets a share of source importance.

2. Model answer

Stack.

3. Model answer

A reference to a key in another table.

4. Model answer

Forward packets between IP networks.

By the end, you can…

• JavaScript can update local interface promptly.

• Server authenticates, authorises and validates record changes.

• Balance speed, privacy, load and security boundaries.

Vocabulary — know these terms

client-side processing — Computing performed in browser/device.

server-side processing — Computing performed by backend.

authentication — Checking claimed identity.

authorisation — Checking what a user may do.

NEW LEARNING

Read, annotate and explain

Browser tasks

JavaScript can update local interface promptly.

Backend tasks

Server authenticates, authorises and validates record changes.

Compare deployment

Balance speed, privacy, load and security boundaries.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A pupil edits a teacher-assigned mark in browser developer tools before submitting a form. How is the system protected?

Browser value is untrusted

Identify the input or diagram and predict the first result.

Server checks permissions

Follow the second stage and explain its importance.

Reject forbidden edit

The server must authenticate the user and enforce edit permissions, ignoring unauthorised mark values. It validates requests and writes to the authoritative database only after checking access.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give two benefits and one limitation of client-side processing. [6]

Question 2

Explain why permission to change assessment results must be checked server-side. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A hub offers instant quiz hints, private reports and offline revision. Assign client/server roles and justify. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Browser validation is sufficient to authorise database changes.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.